

San Jose State University

CS249 - Distributed Systems

San Jose, California



Project Proposal on

Creating a qBitTorrent-like application with persistent data storage.

By

Ricardo A. Miranda

Bach Ngo

To

Dr. Ramin Moazeni, Ph.D.

San Jose, California

October , 2026

Table of Contents

List of Abbreviations i

Chapter 1: Introduction..... 1

 1.1 Problem Statement..... 1

 1.2 History and Background 2

Chapter 2: Research and Approach..... 3

 2.1 Research Objective 3

 2.2 Technical Approach 3

Chapter 3: Requirements and Timeline 5

 3.1 Requirements..... 5

 3.2 Timeline 5

References..... 7

List of Abbreviations

IPFS Interplanetary File System

P2P Peer-to-Peer

DHT Distributed Hash Table

BEP BitTorrent Enhancement Proposal

Chapter 1: Introduction

1.1 Problem Statement

The **DIT! (DIT!)**, a theory that has been gaining traction in the 2020s, especially after the acquisition of Twitter by Elon Musk, is a conspiracy theory that asserts that the activity of the internet is composed mainly of bots [1]. A bot is defined as “an autonomous program on the internet or another network that can interact with systems or users”. In other words, a bot, if programmed correctly, can mimic a user on the internet and portray themselves as a human behind the account. This can lead to bot accounts interacting with normal human accounts and potentially other bot accounts, fueling the idea behind the **DIT!**. This can introduce financial concerns such as content creators experiencing less growth or companies choosing where to advertise. For example, Twitch, a large online streaming platform owned by Amazon, has consistently dealt with view-botting and has given directions on how streamers can handle the situation if they suspect that their channels are being view-botted [2]. View-botting, explained by the twitch moderation team, can be harmful towards streamers because it disallows them to grow their channels organically. This can prevent them from obtaining opportunities in the future. There are also societal concerns such as the spread of misinformation and the skewing of public opinion through bot activity.

This project aims to look at **AI! (AI!)**, **LLM! (LLM!)**, and more specifically, **GPT! (GPT!)**, and what roles they play in the creation of bot accounts, if any. We will also try to create bot accounts of our own to explore the complexities and limitations regarding the creation of bot accounts, as well as the motivations behind them. The main purpose of this project is to see how bot activity can influence the political climate on social media platforms, specifically X, formerly known as Twitter. We will collect data that X allows us to, such as impressions, engagement, and replies.

1.2 History and Background

Bots have regularly been used by companies and government entities on a variety of different platforms, such as phone hot-lines to California's Medical benefits, virtual assistants on all the latest smart phones, and chat bots on company websites with automated responses to help with common questions. They are useful in all activities where human responses would be necessary. Normally, these bots are easy to detect due to their limited automated responses, but with the advancement of technologies such as **ML!** (**ML!**), **LLM!**s, and **GPT!**s, bots are now able to pass the Turing test [3].

The Turing test is a test created by Alan Turing in which a human interrogates two witnesses on a text-only interface. One witness is a human and the other is a machine or a bot imitating a human. The interrogator then decides which witness is the real human. This is also known as imitation game. If a human cannot tell the difference between a human and a bot a significant number of times, then the bot is said to have passed the Turing test. With the introduction of **GPT!**s and their quick advancement, GPT-4 was said to have been judged as human 54% of the time in comparison to the actual human, who were identified 67% of the time [4]. However, just a month after the publication of J. Duan et al [4], the next iteration of GPT technology, GPT-4.5, passed the Turing test [3].

Because social media has a massive reach and millions of users are on social media apps on a daily basis, it is important to be able to detect bots on these platforms, especially when hate speech is seen to increase on platforms like X [5]. Since Elon Musk's purchase of X, hate speech has increased by about 50% through May 2023. In addition to this, the presence of bots likely did not decrease after Musk's purchase [5]. However, other studies have shown a decrease in bot activity from 2016 to 2024 in regards to the time periods leading up to the presidential elections. In their study, K. Debanath, S. Aich, and A. Srizon found that there was a "76% reduction of detected bot prevalence compared to 2016 levels (2.22% vs. 9.7%)" [6]. This does not mean there is less bot activity on the site but instead, bots may be harder to detect.

Chapter 2: Research and Approach

2.1 Research Objective

The goal of this project is to analyze how bots have been detected in previous studies, implement some of those detection methods to the latest and largest dataset, Twibot-22, regarding X posts and accounts, and create our own bot in an attempt to avoid such detection methods to provide better insight as to how big of a problem bots really are.

It is important to note that creating a bot, analyzing the motivations behind creating bots, and following bot activity may involve ethical lapses such as the spread of misinformation. This can be due to many factors, such as deliberately spreading misinformation to push a narrative, known as disinformation. Another factor that is unavoidable, at least in terms of this project, is the risk that a GPT “hallucinates” and returns false information. A hallucination with respect to AI means that the information was presented as fact, even though the information is false. This was a very common occurrence in earlier GPT models, and although hallucinations have seen a significant downward trend since then, hallucinations can and still occur. Another ethical concern is regarding the use of social media apps and the privacy of its users’. Users interact on X on a daily basis and some of them have public accounts. It is important to maintain and respect the user’s privacy, regardless if their accounts are public or not.

2.2 Technical Approach

We will survey a variety of different methods that have been used for bot detection before, such as ML classifiers [7], LLM Bot Guard [4], and **NLP!** (NLP!) techniques [8]. We’ll organize each method based on what detected best and when the detection occurred. Afterwards, we will use the detection methods on the latest dataset, Twibot-22, which is a GitHub repository created by other researchers who are specializing in the field of bot detection [9]. Once that data is consolidated, we

will create our own X bot using the developer options that X allows its users and create an account from there. We will investigate methods on how to automate via GPTs since they now pass the Turing test and can imitate human responses the best. Afterwards, we will revisit the different methods of bot detection and test them against our own bot to see if it circumvents detection and report our findings.

Chapter 3: Requirements and Timeline

3.1 Requirements

For this project, we will need access to LLMs, MLs, and NLP techniques that have been used for bot detection. Some of these models are free to use, such as GPT-4.5 from OpenAI but others will require licenses or subscriptions. We will require access to the Twibot-22 which is given to us free of charge via the researcher's GitHub repository. And lastly, in order to have better engagement metrics on the bot account for X, a subscription for X will be required as many suspected bot accounts are also verified for a boost to engagement.

3.2 Timeline

The timeline for this project will be done over a period of 16 weeks, from the months of August to December of 2026. Since there are four parts to the project, we will split them up into four-week blocks.

The first set four weeks will focus on surveying the different methods and organizing based on best detection element. The second set four weeks will focus on applying those detection methods to the latest dataset, Twibot-22. This period will also involve setting up the environment to properly test on the dataset. The third set of four weeks will involve creating a bot via a GPT, specifically within the first two weeks of this block, and allowing it to interact with the platform for the remaining two weeks. For the last set of four weeks, while still collecting data from our bot, we will apply the methods of bot detection against our own bot with the data it generated. A Gantt chart is presented below to illustrate the timeline.

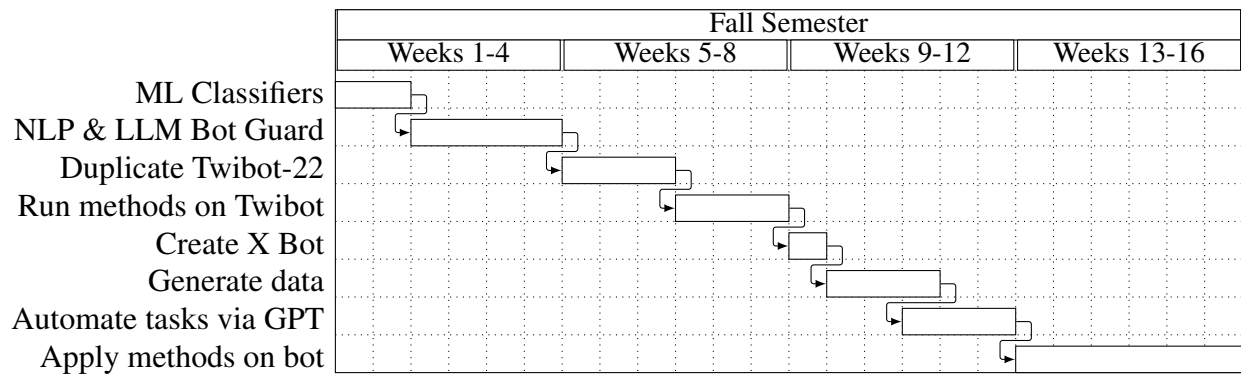


Figure 3.1: Gantt Chart

References

- [1] P. Muzumdar, S. Cheemalapati, S. R. RamiReddy, K. Singh, G. Kurian, and A. Muley, “The dead internet theory: A survey on artificial interactions and the future of social media,” *Asian Journal of Research in Computer Science*, vol. 18, no. 1, pp. 67–73, Jan. 2025, ISSN: 2581-8260. DOI: [10.9734/ajrcos/2025/v18i1549](https://doi.org/10.9734/ajrcos/2025/v18i1549) [Online]. Available: <http://dx.doi.org/10.9734/ajrcos/2025/v18i1549>
- [2] T. Moderation. “How to handle viewership botting and fake engagement.” Accessed: 10-14-2025. [Online]. Available: https://help.twitch.tv/s/article/how-to-handle-view-follow-bots?language=en_US
- [3] C. R. Jones and B. K. Bergen, *Large language models pass the turing test*, 2025. arXiv: [2503.23674](https://arxiv.org/abs/2503.23674) [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2503.23674>
- [4] J. Duan, W. Li, Q. Bai, M. Nguyen, X. Wang, and J. Jiang, “Llm-botguard: A novel framework for detecting llm-driven bots with mixture of experts and graph neural networks,” *IEEE Transactions on Computational Social Systems*, vol. 12, no. 5, pp. 3488–3500, 2025. DOI: [10.1109/TCSS.2025.3545822](https://doi.org/10.1109/TCSS.2025.3545822)
- [5] K. Manke. “Study finds persistent spike in hate speech on x.” Accessed: 10-14-2025. [Online]. Available: <https://vcresearch.berkeley.edu/news/study-finds-persistent-spike-hate-speech-x#:~:text=They%20found%20that%20the%20presence,a%20press%20release%20by%20PLOS.>
- [6] K. Debanath, S. Aich, and A. Y. Srizon, “Analyzing bot activity and political discourse in the 2024 u.s. presidential election: A machine learning approach to misinformation and manipulation,” in *2025 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, 2025, pp. 1–6. DOI: [10.1109/NCIM65934.2025.11160229](https://doi.org/10.1109/NCIM65934.2025.11160229)

- [7] S. Gupta and A. Sinha, “Machine learning classifiers for social media bots detection on twitter using explainable ai,” in *2023 Second International Conference on Informatics (ICI)*, 2023, pp. 1–5. DOI: [10.1109/ICI60088.2023.10421550](https://doi.org/10.1109/ICI60088.2023.10421550)
- [8] H. Mishra, P. Singh, M. Kurien, and S. Javed Quraishi, “Detecting bots in misinformation campaigns: Strategies and societal implications,” in *2025 International Conference on Pervasive Computational Technologies (ICPCT)*, 2025, pp. 917–921. DOI: [10.1109/ICPCT64145.2025.10940500](https://doi.org/10.1109/ICPCT64145.2025.10940500)
- [9] S. Feng et al., *Twibot-22: Towards graph-based twitter bot detection*, 2022. DOI: [10.48550/ARXIV.2206.04564](https://doi.org/10.48550/ARXIV.2206.04564) [Online]. Available: <https://arxiv.org/abs/2206.04564>